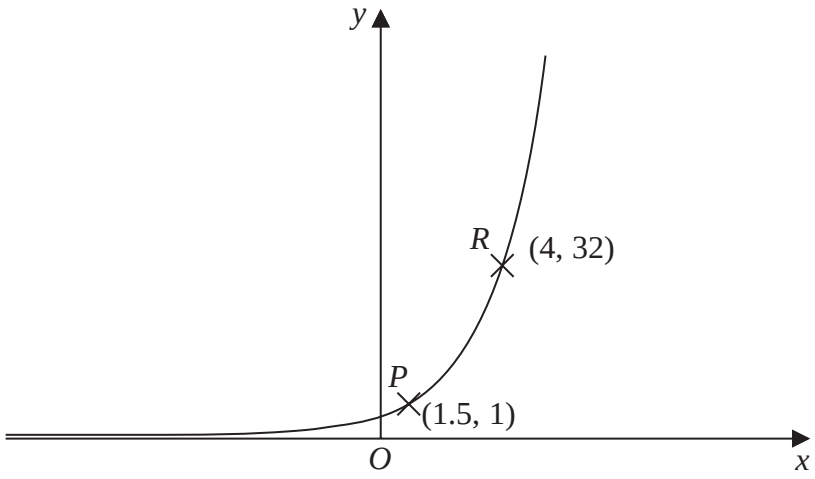
21



The diagram shows a sketch of the graph $y = ab^x$

The curve passes through the point P with coordinates (1.5, 1) and the point R with coordinates (4, 32)

The curve also passes through the point T with coordinates $(-1, k)$

Find the value of k.

$P: 1 = a \times b^{1.5}$

1 mark
Substitute both points.

$R: 32 = a \times b^4$

$Divide: 32 = b^{2.5}$

1 mark
Divide to eliminate a.

$Square: 1024 = b^5 = 4^5$, so $b = 4$

$1 = a \times 4^{1.5} = 8a$, so $a = 1/8$

1 mark
Find b = 4, then a = 1/8.

$At\ x = -1: k = (1/8) \times (1/4) = 1/32$

1 mark
Use x = −1 with a = 1/8 and b = 4.

$k = \frac{1}{32}$

(Total for Question 21 is 4 marks)

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